

RETAIL ANALYTICS • Learner ID: 24CRD05100

Black Friday Retail

Sales & Customer Analytics

Executive Summary & Strategic Recommendations

550,068 transactions • 5,891 customers • \$5.10B in revenue analysed

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Career247- Data Analytics with GenAI Capstone

Methodology: MySQL • Python (pandas / SciPy / stats models) • Power BI

EXECUTIVE OVERVIEW

The Story in One Page

The Black Friday purchases data set consisting of 550,068 transactions among 5,891 customers shows a very concentrated business, from its customer base to its products categories, and even from its target demographic. The concentration, on one hand, is a huge risk; but on the other hand, it is the company's best chance. The high-value customer base, concentrated in a few individuals, generates the lion's share of the revenues and needs to be preserved, while the underutilized customer base, which is dominated by City C, is the largest potential resource for growth; and category breadth has already proven to be a driving factor.

This is why the proposed strategy consists of protecting concentrated high-value revenue, transforming the category behaviour into a repeatable growth model, and adjusting the go-to-market spending for each city according to its unique potential.

\$5.10B TOTAL SALES	550K TRANSACTIONS	5.89K CUSTOMERS	\$9.26K AVG. PURCHASE	62.46% REVENUE FROM TOP 25%
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FINDINGS

Top 5 Insights

- 1

Revenue is concentrated in a small High-Value segment.

The company is highly dependent on 1,473 clients, representing only 25% of the customer base, to bring in 62.46% of its total revenue (\$3.18B).
- 2

Sales are concentrated in a handful of product categories.

Alone category 1 accounts for 37.5% of the total sales (average transaction value \$13,606); the combination of categories 1, 5, and 8 account for almost 72.7%, while the top 5 categories
- 3

City markets show sharply different opportunity profiles.

City B contributes the largest volume (41.52%). However, City C records the biggest number of customers (53.3%) and the smallest percentage of revenue (32.7%) among all cities. This city is therefore the one with the greatest discrepancy between volume and revenue. It is also the one that registers the highest average purchase cost (about \$9,720).
- 4

Category breadth is the strongest behavioural driver of value.

There is a very strong relationship between breadth and total spend ($r \approx 0.734$, $R^2 \approx 0.539$). The High-Value customers make purchases from 13.8 product categories on average compared to 6.3 of Low-Value customers.
- 5

The core customer base is demographically concentrated.

Male customers drive 76.72% of revenue (avg. purchase \$9,438 vs. \$8,735 for female customers); the 26–35 age group contributes 39.9% of revenue from just 34.8% of customers.

STRATEGIC RESPONSE

Recommendations & Priority Matrix

Six recommendations translate the findings above into action, ranked by expected business impact and strength of supporting evidence.

RECOMMENDATION	EVIDENCE	EXPECTED IMPACT	PRIORITY
Launch High-Value retention & loyalty program	Top 25% of customers (1,473) generate 62.46% of revenue (\$3.18B)	Protects the majority of total revenue at lower cost than acquisition	HIGH
Build cross-sell journeys on category breadth	Breadth-spend correlation $r \approx 0.734$; High-Value avg. 13.8 categories vs. 6.3 for Low-Value	Directly actionable lever to move Medium/Low-Value customers up-tier	HIGH
Deploy city-specific go-to-market strategies	City B = 41.52% of sales (scale); City C = highest avg. purchase (~\$9,720) but underpenetrated (53.3% of customers, 32.7% of revenue); City A lowest share (25.83%)	Aligns investment with each market's actual opportunity profile	HIGH
Diversify beyond the core demographic	Male customers = 76.72% of revenue; 26–35 age group = 39.9% of revenue from 34.8% of customers	Reduces concentration risk while protecting near-term revenue	MEDIUM
Segment by occupation for targeted messaging	ANOVA confirms occupation is a statistically significant differentiator of spend	Improves targeting precision beyond generic demographic campaigns	MEDIUM
Deprioritize city-tenure based targeting	No significant correlation between customer tenure in current city and purchase value	Prevents wasted spend on an ineffective segmentation model	LOW

1 • High-Value Retention & Loyalty Program

Build tiered loyalty benefits, early campaign access, dedicated account support, and personalized recommendations around the 1,473 High-Value customers. Retention economics are materially better than acquisition, and even a small increase in churn within this segment carries outsized revenue risk.

2 • Category-Breadth Cross-Sell Program

Use category breadth as a targeting signal: identify Medium- and Low-Value customers purchasing narrowly, and deploy bundled offers and recommendation engines to expand their category footprint. Unlike demographic traits, breadth is directly actionable and has a proven statistical link to spend.

3 • City-Specific Go-to-Market Strategy

Treat each city as a distinct playbook rather than a single national campaign:

- **City B:** protect and scale - the proven volume engine; prioritize retention and supply-chain readiness.
- **City C:** activate on frequency, not discounting - the largest customer base is already there; the gap is engagement, not awareness or price.
- **City A:** stimulate demand and category expansion - smallest share and lowest average purchase, best suited to growth testing.

4 • Demographic Diversification

Focus on preserving the core group (male, 26-35) that contributes to the current income while simultaneously conducting low-cost acquisition tests in the segments that aren't represented well enough – under 18, 55+ age category, and Female customers

5 • Occupation-Based Segmentation

ANOVA testing confirms occupation is a statistically significant differentiator of purchasing behaviour. Shift DMs and Ad creative marketing toward occupation-tailored variants rather than broad demographic-only campaigns.

6 • Deprioritize Tenure-Based Targeting

Customer tenure in current city shows no significant correlation with purchase value. Marketing models built on City Tenure should be retired in favour of the behavioural signals above (value tier, category breadth, city, occupation).

EXECUTION

Measurement & Limitations

Each recommendation should be tracked against a primary KPI so impact is measurable, not assumed:

STRATEGIC LEVER	PRIMARY KPI	EXPECTED IMPACT
High-value retention	High-Value sales share / repeat rate	Protects concentrated revenue base
Cross-sell	Category breadth / avg. purchase	Increases spend depth per customer
City targeting	City sales / avg. purchase	Improves market-level efficiency
Demographic targeting	Avg. purchase / purchase frequency	Improves campaign relevance

A Note on Scope

The dataset does not include campaign cost, discount, promotion, or pre/post-period fields, so promotional ROI cannot be measured directly from this analysis. Recommendations above are therefore built on behavioural performance proxies (value tier, category breadth, city, demographics) rather than direct campaign attribution.